

GroupTrack

User Guide — Version 2.4

April 2026 | Android | For Authorized Testers

GroupTrack is a custom Android app for coordinating multi-vehicle group rides without cellular or internet connectivity. It uses Meshtastic mesh radio hardware to share GPS positions between all vehicles in real time. No phone signal required. No server. No manual radio configuration.

1. USER ROLES

ROLE	DESCRIPTION
ORGANIZER	Creates rides, configures radios, distributes ride configs. Has access to developer settings panel via long press on map.
RIDER	Receives ride config, imports it, connects radio, applies config automatically. Standard user role.

2. SPLASH SCREEN

When GroupTrack launches for the first time or after restart, a branded splash screen displays while the app initializes and waits for mesh nodes. The splash shows the GroupTrack logo, UTV desert background, developer credit, and website. It automatically dismisses when the first mesh node is detected or after a brief initialization period.

3. MAIN MAP SCREEN

The main screen shows a live map with all convoy vehicles plotted as labeled markers. This is the primary operational screen during a ride.

ELEMENT	DESCRIPTION
GroupTrack Header	Tap to open the GroupTrack submenu. Long press to access developer settings (organizer only, password required).
RECORD Button	Top-left red dot. Tap to start GPS track recording. Tap again to access pause/stop options.
Map Settings Panel	Top-right panel — always shows current layer (e.g. SAT) and NET/LOCAL status. Two rows always visible.
Node Markers	Each vehicle shown as a labeled circle. Your cart is larger. Tap any marker to open NODE detail panel.
HUD Overlay	Transparent info panel over the map. Shows convoy or cart stats depending on mode selected.
Button Bar	Fixed bar at bottom: GROUP MY CART HIDE GEAR
Download Progress Chip	Appears above button bar during tile download. Shows tile count, progress, and time remaining.

NOTE: MAP TYPES: SAT = Esri satellite imagery. HYB = Google satellite + roads/trail labels hybrid. TOPO = Esri topographic with terrain. RD = Carto road map with points of interest.

4. BUTTON BAR

BUTTON	DESCRIPTION
GROUP	Shows convoy-wide HUD: all carts, span, lead, tail, active/lost count, channel utilization, GPS interval slider.
MY CART	Shows your vehicle's individual stats: speed, heading, battery, altitude, distance to lead/tail. Also contains lead lo
HIDE	Hides the HUD — map-only view.
GEAR (■)	Opens map settings: map type, zoom controls, search, offline maps, download region.

5. GROUP HUD PANEL

The GROUP HUD displays convoy-wide statistics as a transparent overlay on the left side of the map.

FIELD	DESCRIPTION
Ride Title	Name of the active ride.
UNITS	Total number of cart nodes on the mesh.
ACTIVE	Vehicles heard recently.
LOST	Vehicles that have gone silent.
LEAD	Callsign of the frontmost vehicle (green). After 1/4 mile of movement, the lead is locked and displayed as LEAD L
TAIL	Callsign of the rearmost vehicle (orange). Always derived dynamically.
SPAN	Distance in miles from LEAD to TAIL.
CH%	Channel utilization percentage from radio telemetry. Green <25%, Amber 25-40%, Red >40%.
INT Slider	GPS broadcast interval slider (2-8 seconds). Vertical slider on left side of HUD.

6. MY CART HUD PANEL

FIELD	DESCRIPTION
Callsign	Your convoy callsign.
SPD	Current speed in mph.
HDG	Heading in degrees (0-359).
BAT	Radio battery percentage.
ALT	Altitude in feet.
AHEAD / BEHIND	Distance to nearest vehicles ahead and behind you in the convoy.
TO LEAD / TO TAIL	Distance to front and back of the convoy.
I AM LEAD	Immediately designates your cart as the lead vehicle for the remainder of the ride. No distance accumulatio

RECALC LEAD	Unlocks the current lead assignment. The algorithm runs freely for 1/4 mile of movement, then re-locks to the
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7. NODE HUD — INDIVIDUAL CART DETAIL

Tap any cart marker on the map to open the NODE detail panel for that vehicle.

FIELD	DESCRIPTION
Cart Name	Convoy callsign of the selected cart.
Status	ACTIVE, SIGNAL DROP, or LOST.
SPD / HDG / BAT / ALT	Speed, heading, battery, altitude of the selected cart.
SNR	Signal quality of the last received radio packet.
Position in convoy	Order position, e.g. 3 of 8.
Last seen	Time since last position packet.
RETURN	Returns to full convoy view.
SET AS LEAD	Immediately designates this cart as the lead vehicle. Lock applies instantly — no distance needed. Use a
REMOVE FROM RIDE	Removes cart from active ride roster.

8. MAP SETTINGS PANEL

The map settings panel is in the top-right corner of the map. It is always visible with two rows showing without expanding.

8.1 Map Bar — Two-Row Layout (Always Visible)

The map bar is always visible and provides immediate access to the most-used controls without opening the full panel.

ROW	CONTENTS
Row 1 — Top	▼/▲ chevron (tap to expand/collapse full panel) · MAP label + current layer name · NET/LOCAL toggle switch
Row 2 — Bottom	Layer buttons: SAT HYB TOPO RD — tap any to switch immediately

NOTE: The chevron (▼/▲) on the left of Row 1 is the only expand/collapse trigger. Tapping the layer buttons or the NET/LOCAL switch works without opening the full panel.

8.2 Layer Selection

Four tile source buttons are always visible in Row 2: SAT | HYB | TOPO | RD. Tap any button to switch immediately — no need to expand the full panel. The active source is highlighted blue.

LAYER	DESCRIPTION
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SAT	Esri satellite imagery. Pure overhead satellite — no road or trail labels.
HYB	Google hybrid — satellite imagery with road and trail labels overlaid. Best for navigation on mapped trails.
TOPO	Esri topographic map with terrain contours and elevation data.
RD	Carto road map with points of interest. Best for urban navigation.

8.3 Online / Offline Switch (NET / LOCAL)

The NET/LOCAL toggle switch is always visible in Row 1 of the map bar. No need to expand the panel to switch modes.

STATE	DESCRIPTION
NET (green)	Streaming tiles from internet. Requires WiFi or cell data.
LOCAL (blue)	Serving tiles from device storage. Use before heading into the field.

NOTE: Set to LOCAL before heading into the field. Set to NET when back in connectivity. The label always reflects the current state.

8.4 Download Zoom (Expanded Panel)

Tap the ▼ chevron to expand the full panel. The Download Zoom slider controls the maximum zoom level for tile downloads. Range z16-z18. Higher zoom = more detail but more tiles and larger storage.

8.5 Fly Radius (Expanded Panel)

Slider controlling how close the map zooms when flying to a search result. Range 1-10 miles. 10 miles gives a regional view for selecting a download zone. 1 mile gives a tight trail-level view.

8.6 Location Search (Expanded Panel)

Search for a named area before downloading map tiles. Type a location name (city, park, trail system, or region) and tap FIND AREA. The map flies to the search result and stays there — it will not snap back to your cart position while the search result is displayed.

STEP	DESCRIPTION
1. Type location	Enter city, park name, or trail area. Examples: Zion National Park UT, Gooseberry Mesa, Leeds UT.
2. Tap FIND AREA	Geocoder searches for matching locations. Returns up to 5 results.
3. Select result	If multiple results, tap the correct one from the list. Single results fly directly.
4. Map flies to center	Map zooms to the search result center. A blue pin marks the location. Crosshair lines show lat/lon reference.
5. Zoom and explore	Use pinch-to-zoom to explore trails. The map will not snap back. Touch anywhere on the map to freely pan.
6. Tap DOWNLOAD REGION	Activates the rectangle draw tool. Draw your download area over the search result.

NOTE: If the geocoder cannot find your location, try adding the state abbreviation (e.g. 'Sand Hollow UT'). For precise locations, raw coordinates are supported: 37.44, -113.25.

8.7 Download Region (Expanded Panel)

Tap DOWNLOAD REGION to activate the rectangle draw tool on the map. Draw a box over the area you want to cache for offline use. When you release, a size estimation dialog appears.

DIALOG ELEMENT	DESCRIPTION
Dimensions	Area width x height in miles (e.g. 3.2 mi x 2.1 mi).
Tile count	Number of tiles to download across all zoom levels and all four layers (SAT, HYB, TOPO, RD).
Estimated size	Storage estimate in MB based on average tile size across all layers.
DOWNLOAD button	Starts the download. Downloads all four tile layers simultaneously. Gear panel closes. Toast notification c
CANCEL button	Cancels without downloading.

NOTE: All four tile layers (SAT, HYB, TOPO, RD) are downloaded together for every region. This means all layer types work offline for any area you download. Tiles already on disk are skipped automatically — re-downloading an area only pulls missing layers.

8.8 Download Progress

During download, a progress chip appears above the button bar showing tile count (X of Y tiles) and estimated time remaining across all four layers. The gear panel closes — you can continue using the map while tiles download.

NOTE: Keep app open during download. Background download is not yet supported — closing the app will stop the download. Resume is automatic — tiles already on disk are skipped if you restart a download over the same area.

8.9 Show / Hide Downloaded Areas

The SHOW DOWNLOADED button in the map settings panel displays a blue overlay on the map showing all areas where offline tiles have been downloaded. Use this to identify coverage gaps before heading into the field.

NOTE: The overlay is computed at zoom level 14 for performance. Each blue rectangle represents approximately a 2.5 square mile area. For precise field-level coverage, zoom in past zoom 16 with offline mode active to confirm tile rendering directly.

9. CONVOY SETTINGS

Convoy settings control how GroupTrack tracks the convoy, records the route, and identifies the lead vehicle. These settings are designed around two primary use cases: motorcycle group rides and SXS / UTV convoy rides. Choosing the right combination of settings significantly affects map track clarity and situational awareness.

9.1 Lead Lock

GroupTrack automatically identifies the lead cart by analyzing each vehicle's position in the convoy order. On straight sections this works perfectly. On switchbacks and tight turns at slow speed, position relationships temporarily shift and the algorithm can oscillate between carts, producing spaghetti track lines.

Lead Lock solves this by freezing the lead assignment after the convoy has physically sorted itself into a stable order. The lock fires automatically after 1/4 mile of movement from the RECORD button press — by that distance the convoy is strung out and the lead is clearly established.

BEHAVIOR	DESCRIPTION
Auto-lock after 1/4 mile	After 1/4 mile of movement from RECORD, the lead is locked to whoever the algorithm identified as lead.
LEAD_LOCK_DISTANCE_MILES	Unchangeable constant in ConvoyConfig.kt. Default: 0.25 miles. Can be adjusted by the organizer for different ride types.
Lock persists all ride	Once locked, the lead cart does not change regardless of position oscillation. The track line draws from the lead cart.
SET AS LEAD override	Tap any cart marker on the map, then tap SET AS LEAD in the NODE panel. Immediate lock to that cart.
I AM LEAD override	In MY CART panel. Immediately designates your own cart as lead. Use when you are staging at the front of the ride.
RECALC LEAD	In MY CART panel. Unlocks the current lead, resets the distance accumulator to zero, and re-locks after 1/4 mile.

9.2 Track Recording Mode — Choosing the Right Setting

GroupTrack records GPS track lines during a ride. The Track Recording Mode controls which vehicles draw track lines and what colors are used. Choosing the right mode for your ride type dramatically affects the usefulness of the recorded track.

TRACK ALL (Recommended for Motorcycle Group Rides)

Every cart in the convoy draws its own track line on the map in a distinct color. Each rider's path is individually visible. This is ideal for motorcycle group rides where individual lines, spread, and lateral positioning are meaningful data.

CHARACTERISTIC	DETAIL
Track lines	All carts — each in a distinct assigned color.
Best for	Motorcycle group rides, bicycle groups, any ride where individual paths matter.
Map result	Multiple colored lines showing each rider's exact path. Spread and lateral position visible at a glance.
On tight trails	Multiple colored lines may overlap on single-track sections. This is expected — it shows all riders following the same path.
Setting	GROUP HUD: Track Mode = ALL CARTS. Multicolor enabled.

LEAD ONLY with Black Track (Recommended for SXS / UTV Convoy Rides)

Only the lead cart draws a track line, in solid black. All other carts follow the lead and do not draw independent lines. This produces a single clean authoritative track — the route the convoy actually traveled.

CHARACTERISTIC	DETAIL
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Track lines	Lead cart only — solid black line.
Best for	SXS, UTV, truck, and off-road vehicle convoys where all carts follow the same trail.
Map result	Single clean black track showing the convoy's route. No visual clutter.
Lead Lock importance	Lead Lock is especially important in this mode. If lead oscillates before locking, spaghetti lines appear in t
Setting	GROUP HUD: Track Mode = LEAD ONLY. Multicolor disabled (black track).

NOTE: For the cleanest SXS convoy track: use LEAD ONLY mode with black track, enable Lead Lock, and tap I AM LEAD in MY CART before departure to lock the lead immediately without waiting for 1/4 mile accumulation.

9.3 Convoy Settings Summary — Quick Reference

SETTING	RECOMMENDED VALUE	EFFECT
Track Mode — Motorcycles	ALL CARTS	Every rider's path drawn in individual color. Full route spread visible.
Track Mode — SXS / UTV	LEAD ONLY	Single clean black track from lead cart. No clutter. Best for single-file trail
Multicolor Track	ON for motorcycles / OFF for SXS	ON assigns distinct colors per cart. OFF uses solid black for all tracks.
Lead Lock Distance	0.25 miles (default)	Tunable. Lock fires after this distance from RECORD.
I AM LEAD	Tap before departure	Bypasses distance accumulation. Immediate lock to self.

10. GROUPTRACK MENU

Tap the GroupTrack header at the top of the map to open the submenu. This is the main navigation hub for ride management and radio configuration.

ITEM	DESCRIPTION
Create a Ride	Create a new ride with name, date, and auto-generated encrypted channel. Organizer only.
Transfer Event/Ride Details via Email	Select a ride and send to riders as a .convoy email attachment.
Apply Ride Info to Radio	Configure the connected radio for MASTER baseline or a specific ride.
Restore Prior Config Post Ride	Select and restore a saved radio backup from the archive list.

11. APPLY RIDE INFO TO RADIO

GroupTrack menu → EVENT / RIDE → Apply Ride Info to Radio. Configures the connected radio for use in a convoy.

MODE	DESCRIPTION
MASTER	Resets radio to GroupTrack baseline: US region, LONG_FAST, 903 MHz (channel 5), 60s GPS, Convoy channel, A
RIDE	Applies master baseline plus the selected ride's unique channel name and AES-256 encryption key.

NOTE: V2.4: Channel name, encryption key, and all 38 config fields travel in a single binary install. One reboot. No second stage required. Total process takes approximately 90 seconds.

12. GPS TRACK RECORDING

ACTION	DESCRIPTION
Start Recording	Tap the red RECORD dot (top left of map). Recording begins immediately. Lead lock distance accumulation begins.
Pause Recording	Tap RECORD while recording — select PAUSE from the menu.
Stop and Save	Tap RECORD while recording — select STOP. File save dialog appears. Name your track file.
Export Track	Open Map Settings (expand panel) → Export Tracks. Select files from list, tap EXPORT SELECTED to copy to clipboard.
Track File Format	GPX format. Saved to device storage. Can be imported into mapping apps.

13. TROUBLESHOOTING

PROBLEM	SOLUTION
No vehicles on map	Check radio is connected (green status). Verify all radios use same channel name and PSK. Run Verify Channel.
Yellow lock icon on radio	Encryption not active. Run Apply Radio again — complete the full process.
Spaghetti track lines	Lead oscillating before lock. Use I AM LEAD before departure for instant lock. Or increase LEAD_LOCK_DISTANCE.
Wrong cart locked as lead	Tap RECALC LEAD in MY CART panel. Algorithm re-runs for 1/4 mile and re-locks. Or tap the correct cart.
Map snaps back after zoom	Tap GROUP or MY CART button to re-center intentionally. Any touch on the map disables auto-pan unless zoomed.
Map snaps back after search	Fixed in V2.4. Map stays at search result. If still occurring, tap the search result again.
Map tiles not loading online	Check NET/LOCAL switch in map bar Row 1. If NET, check WiFi or cell data.
Map shows blank tiles offline	No tiles downloaded for this area or zoom level. Use SHOW DOWNLOADED in map settings to see current download.
Only SAT tiles offline, others blank	Re-download the region — all four layers (SAT, HYB, TOPO, RD) must be downloaded together. The first layer determines the zoom level.
Download blocked — too large	Area exceeds download ceiling. Reduce the selected rectangle and try again.
Location search no results	Add state abbreviation (e.g. 'Zion UT'). For precise locations use coordinates: 37.44, -113.25.
Bluetooth disconnects	Set GroupTrack battery to Unrestricted. Settings > Apps > GroupTrack > Battery > Unrestricted.